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Mating interfaces reference actual power/control PCB connector numbers. No extra PCB is implied.

High-voltage input, precharge, choke and PE

文件: 01_hv.kicad_sch

Auxiliary supply, contactors, thermal chain and fan

GND_CTRL

GND_CTRL

文件: 02_aux.kicad_sch

Current-sensor signal harness

GND_CTRL

GND_CTRL

文件: 03_sensor.kicad_sch

Straight-through control harness

文件: 04_harness.kicad_sch

ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED

Sheet: /
File: sic-v3-system.kicad_sch

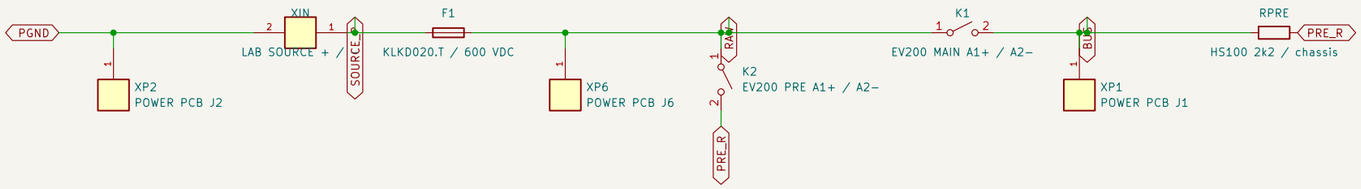
Title: V3 system wiring / off-board assembly

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 1/5

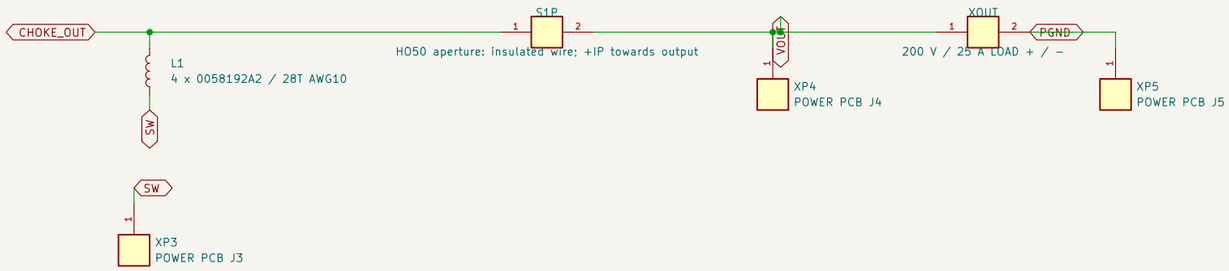
High-voltage input, precharge, choke and PE

S1P is one continuous insulated primary conductor through the sensor aperture; no electrical sensor terminal exists there. K1 and K2 are normally open. Their coil/auxiliary symbols are on sheet 02. RPRE is bolted to the PE thermal plate. Output remains hazardous. Contactors do not provide service isolation; lock out both DC and auxiliary sources. No MOV: controlled source terminal transients <=500 V is an operating restriction; fault-energy coordination remains open.

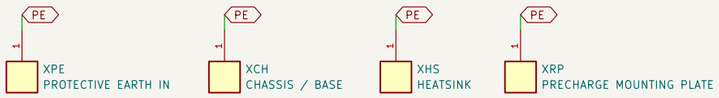
Fused input, parallel main/precharge paths; 360-440 V DC



Chassis choke, insulated primary conductor, hazardous output



Dedicated PE bonds; no signal-ground or power-return bond implied



ENGINEERING PROTOTYPE REVIEW - RELEASE BLOCKED

Sheet: /High-voltage input, precharge, choke and PE/
File: 01_hv.kicad_sch

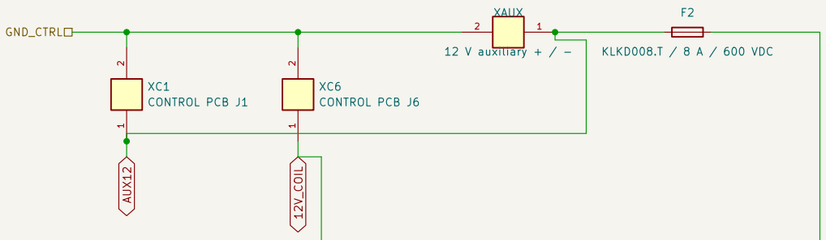
Title: High-voltage input, precharge, choke and PE

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 2/5

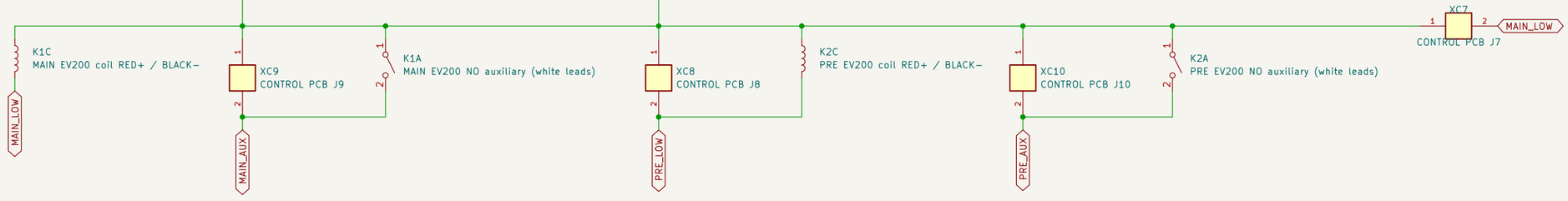
Auxiliary supply, contactors, thermal chain and fan

TS1, TS2 and SSTOP symbols represent contacts closed in the healthy/cold condition; open-on-rise or actuated opens the chain. Actual thermostat contact load: 3.3 V /3.3 mA. Dry-contact reliability and temperature lag require qualification. F2: KLKD008.T in separate L60030M1C + SPL001. Inrush, derating and source/fuse coordination still require qualification. K1/K2 each require up to3.8 A for130 ms. Keep their pickup-current returns out of the electronics input harness. M1 PWM open behavior and lead colors require exact-revision supplier confirmation before wiring.

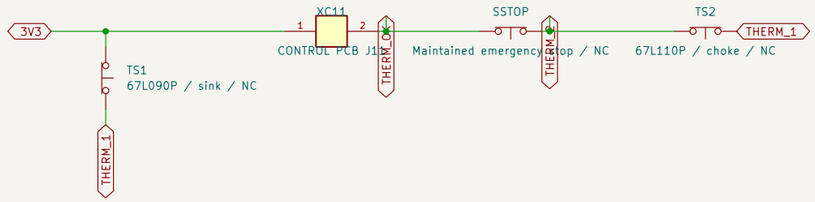
Independent isolated 12 V / 10 A source, separate branch returns



Contactor coils and auxiliary contacts



Fail-open chain; not a certified emergency-stop safety function



12 V fan tach supervised by firmware; PWM lead insulated open



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED

Sheet: /Auxiliary supply, contactors, thermal chain and fan/
File: 02_aux.kicad_sch

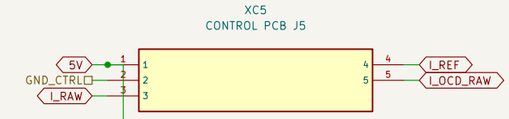
Title: Auxiliary supply, contactors, thermal chain and fan

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 3/5

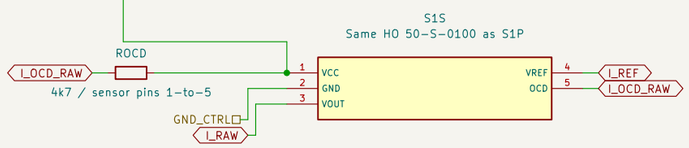
Current–sensor signal harness

ROCD MUST be assembled at the sensor connector. A disconnected harness must pull the PCB I_OCD_RAW low through its own network. Primary insulation, conductor aperture fit, sensing polarity and reference calibration require hardware verification.

Control–board mating connector; 5–wire cable to sensor



SENSOR END: fit ROCD here, beside sensor connector



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED

Sheet: /Current–sensor signal harness/
File: 03_sensor.kicad_sch

Title: Current–sensor signal harness

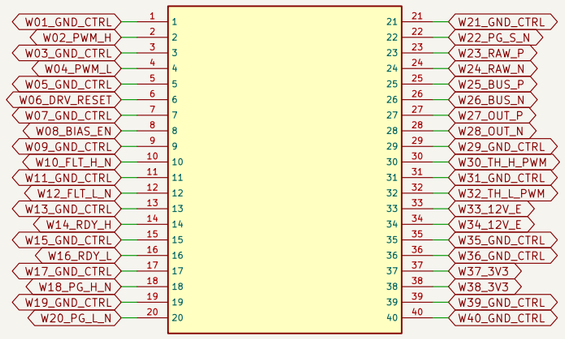
Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 4/5

Straight-through control harness

Each Wxx_SIGNAL net denotes a physical conductor. Same-named grounds/rails are connected on the actual boards. Samtec IDSD cable mates TST headers. Confirm polarisation, pin1, retention and finished-length convention on assembly drawing.

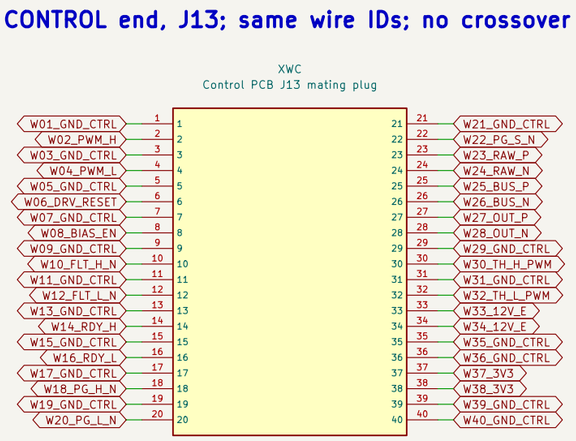
160 mm keyed cable: POWER end, J7; pin n to pin n

XWP
Power PCB J7 mating plug



CONTROL end, J13; same wire IDs; no crossover

XWC
Control PCB J13 mating plug



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED

Sheet: /Straight-through control harness/
File: 04_harness.kicad_sch

Title: Straight-through control harness

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 5/5